A Frobenius-bound algorithm for precedence-constrained SubsetSum with exact target weight

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Editorial note. Editable faithful author material corresponding to official CCC2020 CC BY3.0 publication, article27. One unconditional precedence-constrained SubsetSum/WeightedDAGImplications algorithm mechanism: exact Problem3.1, Definition3.2 Frobenius propertiesP1–P4, complete Lemma3.3 Frobenius gcd criterion and full Bezout/closure/layer/weight induction, Theorem3.4 full recursive algorithm and full correctnessClaim3.5/runtimeClaim3.6 proofs. Original graph/descendant/weight notation from Section2 retained. Exact target weight k and f(k)n^(4sqrt(k)) running time retained, with author f(k)=k^O(k) analysis. No claim of conditional classification or clique/hyperclique lower bound by this item; these are outside this unconditional algorithm theme. All35 official bibliography entries retained. Private native arxiv source is nonexclusive and used only to assist transcription of exactly corresponding official Section3 content; no unlicensed-only text or private native source selected for publication. Official PDF comparison confirms same statements and full proof sequence in pp27:10–13. Original printed four-steps versus listedthree wording, size/weight wording and missing-divides phrase in Claim3.5 are preserved without mathematical correction. Comments replaced with percent placeholders to avoid paragraph break issues. Original claim proof corner typography adapted to standard proof ending. Overview figure is not part of this algorithm proof and is not required. Root handles compilation/database/publication.

Allowed standard prerequisites (editorial).

- Strongly connected component contraction. Original reduction citingTarjan[33] retained.
- Bezout identity and gcd. Original complete author induction using Bezout retained; no recursion into standard number theory.

Theorem (Author Theorem3.4)**.** *We can solve Weighted DAG Implications in time* $f(k)n^{4\sqrt{k}}$.

Weighted DAG Implications asks, given a DAG $G = (V, E)$ with node weights $w : V \to \mathbb{N}$ and parameter $k \in \mathbb{N}$, whether there is a closed set $S \subseteq V$ of weight exactly k, i.e., $w(S) = k$ (author Problem3.1).

All graphs considered in this paper are simple, i.e., we disallow multiple edges and self-loops. If $G = (V, E)$ is a directed graph, we call $S \subseteq V$ a *closed* set if for all $(u, v) \in E$, we have that $u \in S$ implies that $v \in S$. We say that v is a *descendant* of u if v is reachable by a path from u and let $D(u)$ denote the set of descendants of u (including u itself). Analogously, if v is a descendant of u, we call u an *ascendant* of v. We extend the notation naturally to sets $S \subseteq V$ by defining $D(S) \coloneqq \bigcup_{u \in S} D(u)$. For a graph $G = (V, E)$ with node weights $w : V \to \mathbb{N}$ and $S \subseteq V$, we write $w(S) \coloneqq \sum_{v \in S} w(v)$. For any $S \subseteq V$, we let $G[S]$ denote the subgraph of G induced by S, i.e., the subgraph obtained by deleting all vertices in $V \setminus S$ and adjacent edges.

3 Algorithm for Implications

In this section, we give an algorithm for the problem Implications = SAT({IMPL}) that is much faster than brute force and achieves $O(\sqrt{k})$ dependence of k in the exponent n. For convenience, we reduce Implications to the following problem. (Recall that for any graph $G = (V, E)$, we say that $S \subseteq V$ is *closed*, if for all $(u, v) \in E$, we have $u \in S$ implies $v \in S$.)

Problem 3.1 (Weighted DAG Implications)**.** *Given an DAG $G = (V, E)$ with node weights $w : V \to \mathbb{N}$ and parameter $k \in \mathbb{N}$, determine whether there is a closed set $S \subseteq V$ of weight exactly k, i.e., $w(S) = k$.*

The easy reduction works as follows. For each variable x_i, we introduce a corresponding vertex x_i of weight 1 and introduce an edge (x_i, x_j) for every implication constraint $x_i \Rightarrow x_j$ of ϕ. We contract each strongly connected component $C = \{v_1, v_2, \ldots, v_\ell\}$ in G to a single vertex v_C of weight $\sum_{i=1}^{\ell} w(v_i)$ in time $\mathcal{O}(n + m) = \mathcal{O}(n^2)$ [33]. Observe that the resulting graph is a DAG which has a closed set of weight k if and only if ϕ has satisfying assignment of weight k. Recall that for any $v \in V$, we let $D(v)$ denote the set of descendants of v, i.e., the set of nodes reachable from v (including v). As we will formally argue later, by a $f(k)n^{\mathcal{O}(\sqrt{k})}$-time preprocessing it is not difficult to preprocess a Weighted DAG Implications instance into the following form, which we call *Frobenius instance*, as it admits a combinatorial characterization of solvability that is analogous to Schur's bound for the Frobenius coin problem.

Definition 3.2. *A* Frobenius instance with parameter k *is a weighted directed graph $G = (V, E, w)$ with ℓ parts $V = V_1 \cup V_2 \cup \cdots \cup V_\ell$ and weight function $w : V \to \mathbb{N}$ such that the following properties hold:*

(P1) there are weights $w_1, \ldots, w_\ell$ such that $w(v) = w_i$ for all $v \in V_i$ and $i \in [\ell]$.

(P2) for any edge $(u, v) \in E$, we have $u \in V_i$ and $v \in V_j$ for some $\ell \geq i > j \geq 1$,

(P3) for all $i \in [\ell]$, we have $|V_i| \geq k$,

(P4) for all $v \in V$, we have $w(D(v)) \leq \sqrt{k/2}$.

Intuitively, the necessary preprocessing follows from the following arguments: To ensure (P4), note that any weight-k closed set S has at most $\sqrt{2k}$ many vertices $v \in S$ with $w(D(v)) > \sqrt{k/2}$, which we can exhaustively enumerate with $n^{\mathcal{O}(\sqrt{k})}$-time overhead. By suitably arranging remaining nodes among the layers, it is straightforward to ensure (P1), (P2) and additionally that $\ell \leq f(k)$, since by (P4), each node has at most $\mathcal{O}(\sqrt{k})$ descendants. Finally, to ensure (P3), if any part V_i is small (i.e., $|V_i| < k$), we can exhaustively try out including any subset of V_i, introducing an overhead of only $2^{\mathcal{O}(k)}$ per V_i; since $\ell \leq f(k)$, this additional overhead is bounded by $f(k)2^{\mathcal{O}(k)}$. If a Frobenius instance had no edges, then Schur's bound on the Frobenius coin problem implies that it has a solution if and only if $\gcd(w_1, \ldots, w_\ell) \mid k$. We prove that this criterion holds even in the setting of precedence constraints.

Lemma 3.3. *Let G be a Frobenius instance with parameter k. Then G has a closed set of weight k if and only if $\gcd(w_1, \ldots, w_\ell) \mid k$.*

Proof. Since $\gcd(w_1, \ldots, w_\ell) \mid w(S)$ for any $S \subseteq V$, the condition $\gcd(w_1, \ldots, w_\ell) \mid k$ is necessary for G to have a closed set of weight k. We show that this condition is also sufficient via induction on ℓ. In the base case $\ell = 1$, let $S \subseteq V_1 = V$ be an arbitrary subset of k/w_1 vertices (note that by $k/w_1 \leq k \leq |V_1|$, such a set indeed exists). By construction, S has weight $|S|w_1 = k$ and is closed, as G cannot contain any edges. Thus let us assume that the claim holds for all $\ell' \leq \ell - 1$ and consider a Frobenius instance with $d' := \gcd(w_1, \ldots, w_\ell) \mid k$. Let $d := \gcd(w_1, \ldots, w_{\ell-1})$. We may assume that $d \nmid k$; otherwise, already the Frobenius instance $G[V_1 \cup \cdots \cup V_{\ell-1}]$ satisfies the assumption $\gcd(w_1, \ldots, w_{\ell-1}) \mid k$ and we obtain a closed set by inductive hypothesis. Intuitively, we want to use the variables in V_ℓ to reach the target weight k modulo d; then we can reduce to a simpler instance where every weight (including the target weight) is divided by d. Note that we may assume

$$2 \leq d \leq \sqrt{k/2}, \tag{1}$$

where the lower bound follows from $d \nmid k$ and the upper bound follows from $d \leq \min_{i \in [\ell-1]} w_i \leq \sqrt{k/2}$, as $w(v) \leq w(D(v)) \leq \sqrt{k/2}$ for any $v \in V$. Let b be the smallest non-negative integer such that $b \cdot w_\ell \equiv k \pmod{d}$. Such an integer exists and satisfies $b < d$: By Bézout's identity, since $\gcd(w_\ell, d) = d' \mid k$, there are coefficients β, γ such that $\beta w_\ell + \gamma d = k$, and thus any b with $b \equiv \beta \pmod{d}$ achieves the desired congruence. Let $S \subseteq V_\ell$ be an arbitrary subset of size $b < d$; such a set indeed exists as $d \leq \sqrt{k/2} \leq k \leq |V_\ell|$. We observe that S satisfies

$$w(D(S)) \leq \sum_{s \in S} w(D(s)) \leq |S|\sqrt{k/2} \leq d\sqrt{k/2} \leq \frac{k}{2}, \tag{2}$$

where we used (P4) for the second inequality, and (1) for the last inequality. Consider the graph $G' = (V', E')$ obtained as a copy from G from which we delete $V_\ell \cup D(S)$ and define the node weights $w'(v') = w(v)/d$ for any $v \in V \setminus (V_\ell \cup D(S))$. We claim that G' is a Frobenius instance with parameter $k' := (k - w(D(S)))/d$ (observe that k' is indeed integer, as $w(D(S)) \equiv bw_\ell \equiv k \pmod{d}$, and that $k' \geq 0$ by (2)). If this is indeed the case, then by inductive hypothesis G' has a closed set S' with $w'(S') = k'$, since the gcd of the weights w' is 1. Observe that by construction, $D(S) \cup S'$ is a closed set in G of weight

$w(D(S)) + d \cdot w'(S') = w(D(S)) + (k - w(D(S))) = k$, as desired. It remains to prove that G' is indeed a Frobenius instance with parameter k'. First, observe G' has $\ell - 1$ layers $V'_i := V_i \setminus D(S), i \in [\ell - 1]$ and that w' is well defined, as $d \mid w_i$ for all $i \in [\ell - 1]$. Conditions (P1) and (P2) of being Frobenius are fulfilled as G' is a subgraph of G. To see (P3), note that

$$|V'_i| \geq |V_i| - |D(S)| \geq |V_i| - w(D(S)) \geq k - w(D(S)) \geq k'.$$

To see (P4), we observe that by (2) and (1), we have

$$k' = \frac{k - w(D(S))}{d} \geq \frac{k - k/2}{d} = \frac{k}{2d} \geq \frac{k}{d^2}.$$

Thus, for any $v' \in V'$, we obtain

$$w'(D(v')) \leq \frac{w(D(v))}{d} \leq \frac{\sqrt{k/2}}{d} = \sqrt{\frac{k}{2d^2}} \leq \sqrt{k'/2},$$

where we used condition (P4) of G in the second inequality. Thus, G' is indeed a Frobenius instance with parameter k', concluding the claim and thus the proof of our lemma. □

The above criterion is the main technical tool in the algorithmic result of the session. What remains is to show that the instance can be preprocessed in a way that it becomes a Frobenius instance.

Theorem 3.4. *We can solve* Weighted DAG Implications *in time* $f(k)n^{4\sqrt{k}}$.

Proof. Consider the following recursive algorithm, which proceeds in 4 steps: *Step 1:* For every $v \in V$ with $w(D(v)) \geq \sqrt{k/2}$, we return YES if a recursive call determines that $G[V \setminus D(v)]$ has a closed set of weight $k - w(D(v))$; otherwise, we delete v and all its ancestors from G. From now on, G satisfies $w(D(v)) \leq \sqrt{k/2}$ for all $v \in V$. *Step 2:* We construct layers $L_1, \ldots, L_{\sqrt{k/2}}$ by the following iterative process: for every $i = 1, \ldots, \sqrt{k/2}$, we let L_i consists of all vertices in $V \setminus (L_1 \cup \cdots \cup L_{i-1})$ whose outgoing edges end in $L_1 \cup \cdots \cup L_{i-1}$. Note that $L_1, \ldots, L_{\sqrt{k/2}}$ partitions V; in particular, every vertex is included in some L_i, since if there was a vertex $v \in V \setminus (L_1 \cup \cdots \cup L_{\sqrt{k/2}})$, then by construction there exists a path from v containing strictly more than $\sqrt{k/2}$ vertices, leading to the contradiction $w(D(v)) \geq |D(v)| > \sqrt{k/2}$. We observe that each layer L_i can be partitioned into sublayers $L_{i,j}, j \in \{1, \ldots, \sqrt{k/2}\}$ such that each $v \in L_{i,j}$ has weight $w(v) = j$: there can be no vertex of larger weight, as otherwise $w(D(v)) \geq w(v) > \sqrt{k/2}$ yields a contradiction. We consider layers $L_{i,j}$ in increasing lexicographic order of (i, j): If $|L_{i,j}| < k$, then for every $v \in L_{i,j}$, we return YES if a recursive call determines that $G[V \setminus D(v)]$ contains a closed set of size $k - w(D(v))$, and otherwise we delete v and all its ancestors from G. Observe that by the lexicographic ordering, we never delete vertices from already processed layers, so that at the end of the process, each $L_{i,j}$ is either empty or contains at least k vertices. *Step 3:* We let $V_1, \ldots, V_\ell$ be an enumeration of all non-empty sublayers $L_{i,j}$ by the lexicographic order on (i, j) so that any vertex $v \in V_i$ has only edges to vertices in $V_1 \cup \cdots \cup V_{i-1}$. Observe that by construction, this yields a Frobenius instance. Let $w_1, \ldots, w_\ell$ be the weights of the Frobenius instance. We return YES if $\gcd(w_1, \ldots, w_\ell) \mid k$ and NO otherwise. Using Lemma 3.3, the correctness of the algorithm is easy to see.

Claim 3.5. *The above algorithm is correct.*

Proof. If the algorithm returns YES, indeed there is a closed set of size k: If we return YES in Steps 1 or 2, we have found a vertex v and a closed set S' in $G[V \setminus D(v)]$ of size $k - w(D(v))$, which yields a closed set $S' \cup D(v)$ in G of size k, as desired. Otherwise, we have arrived at a Frobenius instance and returned YES since $\gcd(w_1, \ldots, w_\ell) \mid k$, which implies that G has a closed set of size k by Lemma 3.3. Conversely, fix a closed set S of size k, and we show that the algorithm returns YES: If S contains a vertex v investigated in Steps 1 or 2, then the recursive call to $G[V \setminus D(v)]$ (for the first such vertex v) will find a solution of size $|S| - w(D(v))$ (note that $D(v) \subseteq S$ if $v \in S$). Otherwise, we have arrived at a Frobenius instance which must satisfy $\gcd(w_1, \ldots, w_\ell)$ by Lemma 3.3, and we return YES. □

Finally, we need to bound the running time of the recursive algorithm. The analysis relies on the observation that the algorithm makes at most n recursive calls with a parameter decrease of at least $\sqrt{k/2}$, and at most $O(k^2)$ recursive calls with a parameter decrease of one.

Claim 3.6. *The above algorithm can be implemented in time $f(k)n^{4\sqrt{k}}$.*

Proof. Let U be the set of vertices of small layers ($|L_{i,j}| < k$) considered in Step 2. We observe that the above algorithm can be implemented recursively with the following recurrence on its running time $T(n, k)$ on instances with n vertices and parameter k.

$$T(n,k) \leq \sum_{v \in V, w(D(v)) \geq \sqrt{k/2}} T(n, k - w(D(v))) + \sum_{u \in U} T(n, k - w(D(u))) + \mathcal{O}(n^2)$$

We claim by induction on k that this yields a bound of $T(n, k) \leq f(k)n^{4\sqrt{k}}$ for some $f(k) = k^{\mathcal{O}(k)}$. It is not difficult to see that for $k \leq 2$, we can solve the problem in time $\mathcal{O}(n^2) = \mathcal{O}(n^{4\sqrt{k}})$, yielding the base case. For $k \geq 3$, we thus obtain the following bound, using that in Step 2, we process less than k vertices for each "small" sublayer $L_{i,j}, 1 \leq i, j \leq \sqrt{k/2}$, i.e., $|U| \leq k(k/2) = k^2/2$,

$$\begin{aligned} T(n,k) &\leq \mathcal{O}(n \cdot f(k - \sqrt{k/2})n^{4\sqrt{k-\sqrt{k/2}}} + k^2 f(k-1)n^{4\sqrt{k-1}} + n^2) \\ &\leq (f(k)/2)(n^{4\sqrt{k-\sqrt{k/2}}+1} + n^{4\sqrt{k}}) \leq f(k)n^{4\sqrt{k}}, \end{aligned}$$

where the second bound follows from choosing $f(k) = k^{\mathcal{O}(k)}$ large enough to ensure $k^2 f(k-1) \leq f(k)/2$ and the last bound follows from the observation that $4\sqrt{k - \sqrt{k/2}} + 1 \leq 4\sqrt{k}$ if and only if

$$\begin{aligned} & \left(4\sqrt{k - \sqrt{k/2}} + 1\right)^2 \leq 16k \\ \iff \quad & 16(k - \sqrt{k/2}) + 8\sqrt{k - \sqrt{k/2}} + 1 \leq 16k \\ \iff \quad & 8\sqrt{k - \sqrt{k/2}} + 1 \leq 16\sqrt{k/2}, \end{aligned}$$

where the last inequality holds since $8\sqrt{k} + 1 \leq 16\sqrt{k/2}$ as $k \geq 3$. □

Claims 3.5 and 3.6 show the correctness of our algorithm for Weighted DAG Implications. By the reduction described at the beginning of the section, a similar algorithm follows for Implications. □

References

[1] Amir Abboud, Arturs Backurs, Karl Bringmann, and Marvin Künnemann. Fine-grained complexity of analyzing compressed data: Quantifying improvements over Decompress-and- Solve. In Proc. 58th IEEE Annual Symposium on Foundations of Computer Science (FOCS 2017), pages 192–203, 2017. doi:10.1109/FOCS.2017.26.

[2] Amir Abboud, Arturs Backurs, and Virginia Vassilevska Williams. If the current clique algorithms are optimal, so is valiant's parser. SIAM J. Comput., 47(6):2527–2555, 2018. doi:10.1137/16M1061771.

[3] Amir Abboud, Karl Bringmann, Holger Dell, and Jesper Nederlof. More consequences of falsifying SETH and the Orthogonal Vectors conjecture. In Proc. 50th Annual ACM SIGACT Symposium on Theory of Computing (STOC 2018), pages 253–266, New York, NY, USA, 2018. ACM. doi:10.1145/3188745.3188938.

[4] Noga Alon, Raphael Yuster, and Uri Zwick. Color-coding. J. ACM, 42(4):844–856, 1995. doi:10.1145/210332.210337.

[5] Enric Boix-Adserà, Matthew Brennan, and Guy Bresler. The average-case complexity of counting cliques in Erdős-Rényi hypergraphs. In Proc. 60th IEEE Annual Symposium on Foundations of Computer Science (FOCS 2019), pages 1256–1280, 2019. doi:10.1109/FOCS. 2019.00078.

[6] Édouard Bonnet, László Egri, and Dániel Marx. Fixed-Parameter Approximability of Boolean MinCSPs. In Piotr Sankowski and Christos Zaroliagis, editors, Proc. 24th Annual European Symposium on Algorithms (ESA 2016), volume 57 of Leibniz International Proceedings in Informatics (LIPIcs), pages 18:1–18:18, Dagstuhl, Germany, 2016. Schloss Dagstuhl–Leibniz- Zentrum fuer Informatik. doi:10.4230/LIPIcs.ESA.2016.18.

[7] Alfred Brauer. On a problem of partitions. American Journal of Mathematics, 64(1):299–312, 1942.

[8] Karl Bringmann. A near-linear pseudopolynomial time algorithm for subset sum. In Proc. 28th Annual ACM-SIAM Symposium on Discrete Algorithms (SODA 2017), pages 1073–1084, 2017. doi:10.1137/1.9781611974782.69.

[9] Karl Bringmann, Nick Fischer, and Marvin Künnemann. A fine-grained analogue of Schaefer's theorem in P: dichotomy of $\exists^k\forall$-quantified first-order graph properties. In Proc. 34th Computational Complexity Conference (CCC 2019), pages 31:1–31:27, 2019. doi:10.4230/LIPIcs.CCC.2019.31.

[10] Karl Bringmann, Allan Grønlund, and Kasper Green Larsen. A dichotomy for regular expression membership testing. In Chris Umans, editor, Proc. 58th IEEE Annual Symposium on Foundations of Computer

Science (FOCS 2017), pages 307–318. IEEE Computer Society, 2017. doi:10.1109/FOCS.2017.36.

[11] Andrei A. Bulatov. A dichotomy theorem for nonuniform CSPs. In Proc. 58th IEEE Annual Symposium on Foundations of Computer Science (FOCS 2017), pages 319–330, 2017. doi:10.1109/FOCS.2017.37.

[12] Andrei A. Bulatov and Dániel Marx. Constraint satisfaction parameterized by solution size. SIAM J. Comput., 43(2):573–616, 2014. doi:10.1137/120882160.

[13] Nofar Carmeli and Markus Kröll. On the enumeration complexity of unions of conjunctive queries. In Dan Suciu, Sebastian Skritek, and Christoph Koch, editors, Proc. 38th ACM SIGMOD-SIGACT-SIGAI Symposium on Principles of Database Systems (PODS 2019), pages 134–148. ACM, 2019. doi:10.1145/3294052.3319700.

[14] Jianer Chen, Benny Chor, Mike Fellows, Xiuzhen Huang, David W. Juedes, Iyad A. Kanj, and Ge Xia. Tight lower bounds for certain parameterized NP-hard problems. Inf. Comput., 201(2):216–231, 2005. doi:10.1016/j.ic.2005.05.001.

[15] Jianer Chen, Xiuzhen Huang, Iyad A. Kanj, and Ge Xia. Strong computational lower bounds via parameterized complexity. J. Comput. Syst. Sci., 72(8):1346–1367, 2006. doi: 10.1016/j.jcss.2006.04.007.

[16] Jianer Chen, Iyad A. Kanj, and Ge Xia. Improved upper bounds for vertex cover. Theor. Comput. Sci., 411(40-42):3736–3756, 2010. doi:10.1016/j.tcs.2010.06.026.

[17] Nadia Creignou. A dichotomy theorem for maximum generalized satisfiability problems. J. Comput. Syst. Sci., 51(3):511–522, 1995. doi:10.1006/jcss.1995.1087.

[18] Nadia Creignou, Sanjeev Khanna, and Madhu Sudan. Complexity classifications of boolean constraint satisfaction problems. SIAM, 2001. doi:10.1137/1.9780898718546.

[19] Friedrich Eisenbrand and Fabrizio Grandoni. On the complexity of fixed parameter clique and dominating set. Theor. Comput. Sci., 326(1-3):57–67, 2004. doi:10.1016/j.tcs.2004.05.009.

[20] Tomás Feder and Moshe Y. Vardi. The computational structure of monotone monadic SNP and constraint satisfaction: A study through datalog and group theory. SIAM J. Comput., 28(1):57–104, 1998. doi:10.1137/S0097539794266766.

[21] Jean Gallier. The Frobenius coin problem. Upper bounds on the Frobenius number, 2014.

[22] Sanjeev Khanna, Madhu Sudan, Luca Trevisan, and David P. Williamson. The approximability of constraint satisfaction problems. SIAM J. Comput., 30(6):1863–1920, 2000. doi:10.1137/ S0097539799349948.

[23] Konstantinos Koiliaris and Chao Xu. Faster pseudopolynomial time algorithms for subset sum. ACM Trans. Algorithms, 15(3):40:1–40:20, 2019. doi:10.1145/3329863.

[24] Stefan Kratsch, Dániel Marx, and Magnus Wahlström. Parameterized complexity and kernelizability of max ones and exact ones problems. TOCT, 8(1):1:1–1:28, 2016. doi: 10.1145/2858787.

[25] Andrei A. Krokhin and Dániel Marx. On the hardness of losing weight. ACM Trans. Algorithms, 8(2):19:1–19:18, 2012. doi:10.1145/2151171.2151182.

[26] Bingkai Lin. The parameterized complexity of the k-biclique problem. J. ACM, 65(5):34:1– 34:23, 2018. doi:10.1145/3212622.

[27] Andrea Lincoln, Virginia Vassilevska Williams, and Ryan Williams. Tight hardness for shortest cycles and paths in sparse graphs. In Proc. 29th Annual ACM-SIAM Symposium on Discrete Algorithms (SODA 2018), pages 1236–1252, Philadelphia, PA, USA, 2018. Society for Industrial and Applied Mathematics.

[28] Dániel Marx. Parameterized complexity of constraint satisfaction problems. Computational Complexity, 14(2):153–183, 2005. doi:10.1007/s00037-005-0195-9.

[29] Jaroslav Nešetřil and Svatopluk Poljak. On the complexity of the subgraph problem. Com- mentationes Mathematicae Universitatis Carolinae, 026(2):415–419, 1985.

[30] Jorge L. Ramirez Alfonsin. The diophantine Frobenius problem. Oxford University Press, Oxford, 2005.

[31] Thomas J. Schaefer. The complexity of satisfiability problems. In Proc. 10th Annual ACM Symposium on Theory of Computing (STOC 1978), pages 216–226, 1978. doi:10.1145/ 800133.804350.

[32] Jeanette P. Schmidt and Alan Siegel. The spatial complexity of oblivious k-probe hash functions. SIAM J. Comput., 19(5):775–786, 1990. doi:10.1137/0219054.

[33] Robert Endre Tarjan. Depth-first search and linear graph algorithms. SIAM J. Comput., 1(2):146–160, 1972. doi:10.1137/0201010.

[34] Dmitriy Zhuk. A proof of CSP dichotomy conjecture. In Proc. 58th IEEE Annual Symposium on Foundations of Computer Science (FOCS 2017), pages 331–342, 2017. doi:10.1109/FOCS. 2017.38.

[35] Dmitriy Zhuk and Barnaby Martin. QCSP monsters and the demise of the Chen conjecture. CoRR, abs/1907.00239, 2019. arXiv:1907.00239.